

1. What is Data Science? Explain its primary objective.

Answer:

Data Science is the process of collecting, analyzing, and interpreting data to gain useful insights.

Primary Objective: To help make better decisions using data.

2. Why is Data Science important in today's digital world? Mention any four reasons.

Answer:

- Helps in better decision-making.
- Predicts future trends.
- Improves business performance.
- Automates tasks using AI.

3. List any four benefits of Data Science in business and society.

Answer:

- Better customer service.
- Increased business profits.
- Improved healthcare.
- Fraud detection.

4. Mention four real-world applications of Data Science.

Answer:

- Healthcare
- Banking
- E-commerce
- Transportation

5. What are the different facets of data? Briefly explain each.

Answer:

- **Structured Data:** Organized in tables.
- **Semi-Structured Data:** Partly organized (XML, JSON).
- **Unstructured Data:** No fixed format (images, videos).
- **Metadata:** Data about data.

6. Differentiate between structured, semi-structured, and unstructured data with suitable examples.

Answer:

Type	Example
Structured	Student database
Semi-Structured	XML, JSON
Unstructured	Images, videos, emails

7. What is Big Data? State its key characteristics (5 Vs).

Answer:

Big Data is very large and complex data that cannot be processed by traditional tools.

5 Vs:

- Volume
- Velocity
- Variety
- Veracity
- Value

8. Explain the Big Data ecosystem. Name any four technologies used in it.

Answer:

The Big Data ecosystem includes tools for storing, processing, and analyzing large data.

Technologies:

- Hadoop
- Spark
- Hive
- HBase

9. What are the major phases of the Data Science process?

Answer:

1. Data Collection

2. Data Cleaning
3. Data Analysis
4. Model Building
5. Model Evaluation

10. What is data retrieval? Mention two common sources from which data can be retrieved.

Answer:

Data retrieval means collecting data from different sources.

Sources:

- Databases
- Websites/APIs

11. Why is data collection considered an important step in the Data Science process?

Answer:

Because accurate and relevant data leads to better analysis and results.

12. What is data cleansing? List any four common data-cleaning techniques.

Answer:

Data cleansing is the process of removing errors and improving data quality.

Techniques:

- Remove duplicates
- Handle missing values
- Correct errors
- Remove outliers

13. What is data integration? Why is it necessary in Data Science?

Answer:

Data integration combines data from different sources into one dataset.

Need: It improves accuracy and provides complete information.

14. Explain data transformation with suitable examples.

Answer:

Data transformation changes data into a useful format.

Example: Converting text into numbers or changing date formats.

15. What is data analysis? Mention any four techniques used for data analysis.

Answer:

Data analysis is the process of examining data to find useful information.

Techniques:

- Classification
- Regression
- Clustering
- Visualization

16. Differentiate between descriptive, predictive, and prescriptive analytics.

Answer:

Type	Purpose
Descriptive	What happened?
Predictive	What may happen?
Prescriptive	What should be done?

17. What is meant by building a model in Data Science? Mention the basic steps involved.

Answer:

Model building means creating a machine learning model to make predictions.

Steps:

- Select algorithm
- Train model
- Test model
- Improve model

18. Why is model evaluation important? Name any four evaluation metrics used for classification models.

Answer:

Model evaluation checks how well the model performs.

Metrics:

- Accuracy
- Precision
- Recall
- F1-Score

19. Why is presenting findings an essential part of the Data Science process? Mention any four visualization tools or techniques.**Answer:**

It helps people understand the results and make decisions.

Tools/Techniques:

- Bar Chart
- Pie Chart
- Line Chart
- Tableau

20. How are Data Science models deployed into real-world applications? Give any four examples of Data Science applications.**Answer:**

Models are deployed through web applications, mobile apps, or cloud services.

Applications:

- Recommendation systems
- Fraud detection
- Disease prediction
- Weather forecasting